STI WEST NEGROS UNIVERSITY

College of Information and Communications Technology

Internship Narrative Report: IT Infrastructure Support and Systems Deployment

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A 480-Hour On-the-Job Training Practicum Narrative for Bachelor of Science in Information Technology

Final Practicum Narrative Report

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Acknowledgments

I extend my sincerest gratitude to Almighty God for the perseverance, wisdom, and health granted throughout this practicum journey. My heartfelt appreciation goes to the academic leadership and faculty of STI West Negros University for imparting essential foundational knowledge in software engineering, database administration, and professional ethics.

Special recognition is offered to Horizon Tech Solutions Inc. for welcoming me into their Systems Engineering division, providing invaluable industry mentorship, and trusting me with real-world enterprise infrastructure responsibilities.

1. Company Profile and Host Establishment

Horizon Tech Solutions Inc., established in 2018 in Bacolod City, is an enterprise managed service provider (MSP) specializing in cloud migrations, network administration, and custom enterprise web portals. As an intern, I was assigned to the Tier 2 Systems and Network Administration squad.

The department manages hybrid infrastructure comprising on-premises Linux virtualization hosts and cloud-hosted database clusters serving educational and commercial clients throughout Western Visayas.

2. Summary of Weekly Accomplishments

Over the 480-hour practicum duration, technical responsibilities advanced from initial hardware diagnostics to autonomous server maintenance and database migrations.

Table 1. 10-Week Practicum Milestone and Technical Activity Log

Practicum Phase	Assigned Technical Tasks	Core Competencies Mastered
Weeks 1–2: Orientation	Workstation imaging, Active Directory user provisioning	Windows Server AD, Sysprep, Cat6 cabling
Weeks 3–4: Infrastructure	VLAN configuration, managed switch deployments	Cisco IOS commands, Subnetting, QoS rules
Weeks 5–6: Web Services	Linux server updates, Nginx reverse proxy configuration	Ubuntu Server, SSL Certbot, Bash scripting
Weeks 7–8: Database Ops	PostgreSQL automated backup scripts, query tuning	SQL indexing, pg_dump automation, Cron jobs
Weeks 9–10: Capstone	Internal asset tracking inventory web portal	HTML/CSS/JS, PHP/MySQL,

Project	deployment	Docker containers

KEY INDUSTRY ACHIEVEMENT: Architected an automated bash backup routine that reduced weekly database maintenance downtime from 45 minutes to under 8 minutes across 6 client instances.

3. Deep Technical Workflow: Server Hardening & Automation

During Weeks 5 and 6, our team conducted a comprehensive vulnerability audit on enterprise web application servers. I automated SSH key management and disabled root password logins across 14 virtual machines. Furthermore, I authored custom shell scripts configuring UFW firewall rate limits, effectively dampening brute-force authentication attacks.

4. Challenges Encountered and Problem-Solving Strategies

During Week 6, an unannounced SSL certificate expiration interrupted an enterprise customer's public API endpoint. Working under high-pressure live production conditions, I collaborated with senior engineers to troubleshoot DNS challenges, execute Certbot renewals, and establish automated cron job alerts to prevent recurrence.

5. Self-Evaluation, Insights, and Professional Growth

This practicum underscored that academic theory must be paired with operational resilience. Technical ability is only half the battle; transparent communication, meticulous documentation, and calm problem-solving under tight deadlines are what truly distinguish a reliable IT professional.

Moving forward into my final college year, I plan to pursue specialized certifications in Linux Systems Administration and AWS Cloud Solutions.

6. Practicum Competency Assessment Matrix

Table 2. Final Institutional Competency Ratings by Industry Supervisor

Core Evaluation Domain	Target Performance Standard	Achieved Supervisor Rating
Workplace Ethics & Attendance	Punctuality, dress code compliance, and confidentiality	Outstanding (100%)
Systems Administration	Autonomous Linux maintenance, AD user	Exceeds Standards (96%)

	provisioning	
Problem Analysis & Initiative	Independent root-cause analysis during live outages	Outstanding (98%)
Documentation & Teamwork	Clear technical guides and cross-team communication	Exceeds Standards (95%)

References & Industry Guidelines

Commission on Higher Education. (2017). Policies, standards, and guidelines for the Bachelor of Science in Information Technology (BSIT) program (CHED Memorandum Order No. 25).

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